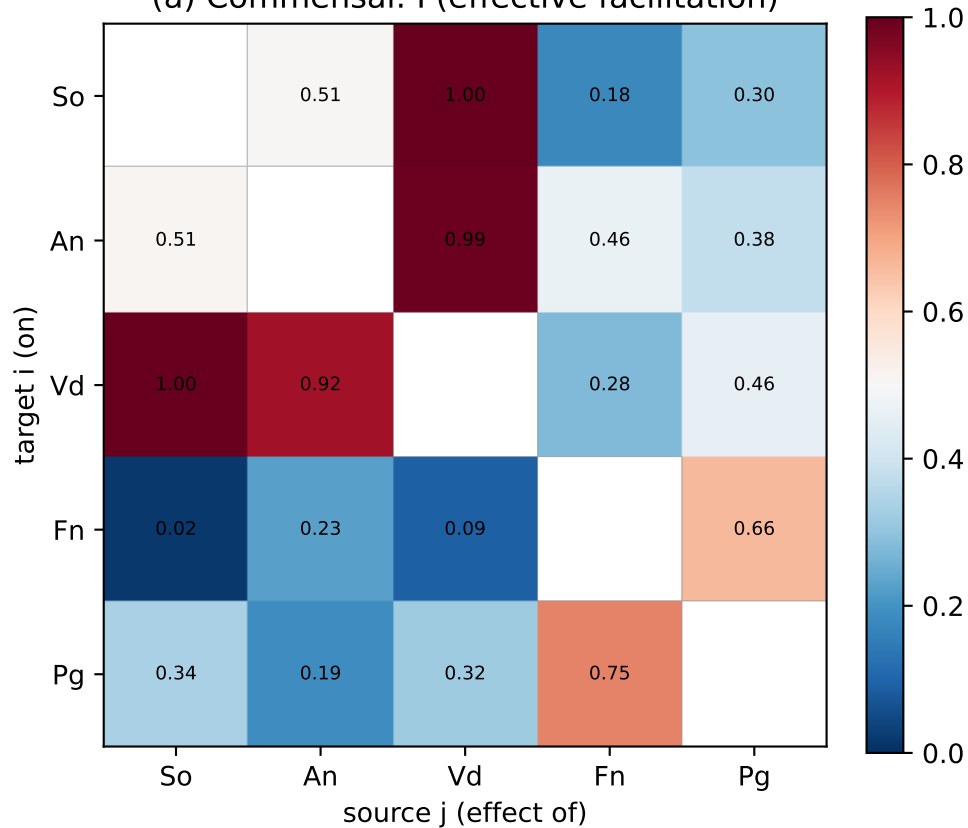
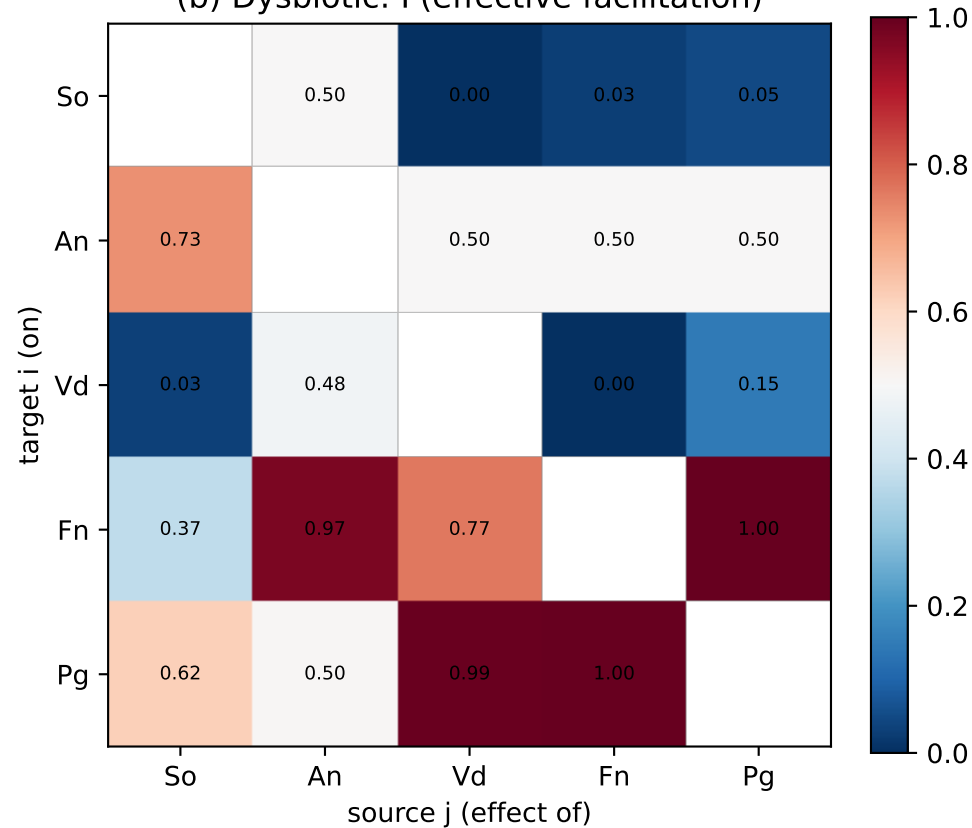
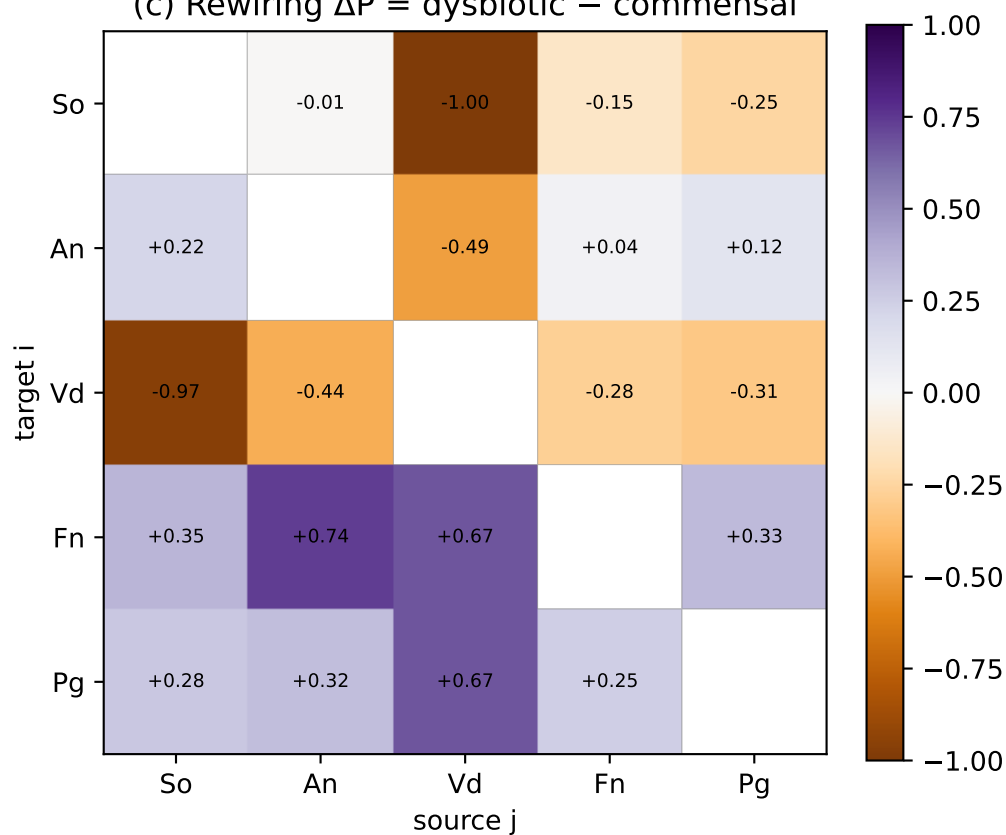


(a) Commensal: P(effective facilitation)



(b) Dysbiotic: P(effective facilitation)

(c) Rewiring $\Delta P = \text{dysbiotic} - \text{commensal}$ 

(d) Pg network role: commensal(CS,CH) vs dysbiotic(DS,DH)

